

Friedrichs Aharonov–Bohm Dynamics: Full Scattering, Heat Kernel, and Determinant Obstructions

HCS-C383 theorem and reproducibility package

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Abstract

We give a single-convention audit of the Friedrichs Aharonov–Bohm Hamiltonian on the punctured plane. The angular Hankel transform yields the complete absolutely continuous spectrum, and the free wave operators give every scattering phase. Abel summation preserves both the forward delta and the principal-value part. Weber’s integral produces a locally convergent heat kernel with an explicit angular tail bound. The global heat semigroup and, at noninteger flux, the scattering correction are noncompact. Two exhaustive channel windows even give different finite determinant limits. Integer gauge covariance classifies position-preserving physical time reversal, while a reflection-composed antiunitary exists at every flux. Exact finite receipts and independent spectral quadrature audit the formulas. The classical results are credited as such; the package advance is their integrated domain, distribution, symmetry, and determinant-boundary verification. This natural scattering system fails the Route-A arithmetic and primitive-orbit gates.

中文摘要

本文在固定磁通符号与物理时间下统一验证弗里德里希斯阿哈罗诺夫玻姆动力学。角向汉克尔变换给出完整纯绝对连续谱，贝塞尔入射出射渐近产生全部散射相位，阿贝尔求和保留前向分布与主值项。韦伯积分进一步给出局部热核及显式角通道尾界，但全空间热半群并不紧，因此不存在通常的有限热迹。非整数磁通下散射修正也不紧，而且两种都穷尽整数通道的截断方式得到不同的乘积极限，揭示普通行列式解释的缺口。整数规范协变把保持空间点的物理时间反演限定在整数和半整数磁通；复合空间反射的反酉对称则适用于所有磁通，两种对称不能混为一谈。精确有限记录、独立谱积分与敌对变异检验共同审查这些约定。经典公式不作为新发现，包内进展是对域、全分布、热核、对称与行列式边界的完整贯通；路线甲仍在算术入口停止。

Keywords: magnetic scattering; Friedrichs domain; heat kernel; gauge covariance; time reversal; noncompactness; Route A

关键词：磁性散射；弗里德里希斯域；热核；规范协变；时间反演；非紧性；路线甲

1 One operator and one boundary condition

On $\mathcal{H} = L^2(\mathbb{R}^2, r dr d\theta)$, set $h = 2M = 1$ and freeze the nonnegative form, initially on smooth functions compactly supported away from zero,

$$q_\beta[f] = \int_0^\infty \int_0^{2\pi} \left(|\partial_r f|^2 + \frac{|(-i\partial_\theta - \beta)f|^2}{r^2} \right) r d\theta dr. \quad (1)$$

Its closed Friedrichs extension defines the self-adjoint H_β . The main flux representative is $0 \leq \beta < 1$; integer translates will only be used for gauge comparisons. No spin, contact interaction, alternate self-adjoint extension, finite-radius flux tube, or confining field enters. This choice is material: an unqualified singular differential expression does not fix its quantum dynamics.

Let $e_m(\theta) = (2\pi)^{-1/2} e^{im\theta}$ and $v_m = |m - \beta|$. Angular Parseval gives the orthogonal form sum

$$q_\beta = \bigoplus_{m \in \mathbb{Z}} q_{v_m}, \quad q_v[u] = \int_0^\infty \left(|u'|^2 + \frac{v^2}{r^2} |u|^2 \right) r dr. \quad (2)$$

Finite angular sums and smooth compact radial approximants form a core of this sum: Fourier truncation converges in both norm and form energy, and the radial closure then approximates each of finitely many components.

Theorem 1 (Domain and complete spectral representation). *The Friedrichs operator decomposes as*

$$H_\beta = \bigoplus_{m \in \mathbb{Z}} h_{v_m}, \quad h_v = -\partial_r^2 - r^{-1} \partial_r + v^2 r^{-2}. \quad (3)$$

The order- ν Hankel map

$$(\mathcal{J}_\nu u)(k) = \int_0^\infty J_\nu(kr)u(r)r \, dr \quad (4)$$

is unitary from $L^2(r \, dr)$ onto $L^2(k \, dk)$ and transforms h_ν to multiplication by k^2 . In particular,

$$D(h_\nu) = \{u : k^2 \mathcal{J}_\nu u \in L^2(k \, dk)\}. \quad (5)$$

The full operator has purely absolutely continuous spectrum $[0, \infty)$, with no point or singular continuous spectrum.

Proof. The form-core argument proves (3). For $0 < \nu < 1$, the two zero-energy branches are r^ν and $r^{-\nu}$. The latter has divergent form energy and its coefficient is removed in the Friedrichs realization. For $\nu = 0$, the logarithmic branch is excluded but the regular constant branch is allowed. Orders $\nu \geq 1$ are limit point at zero, and infinity is limit point in every channel. Thus these are precisely the regular Bessel realizations diagonalized by the Hankel Plancherel and inversion theorem [3]. The transform is onto and gives (5); it is not a finite list of formal eigenfunctions. Multiplication by k^2 under the non-atomic measure $k \, dk$ has the claimed spectral types, and its countable orthogonal sum preserves them. $\square$

Lemma 1 (The zero-flux puncture). H_0 is the ordinary free plane Laplacian.

Proof. For $0 < \epsilon < 1$, take a radial cutoff equal to zero at $r \leq \epsilon^2$, equal to one at $r \geq \epsilon$, and linear in $\log r$ between those radii. Its Dirichlet energy is $2\pi/|\log \epsilon|$. Multiplying a fixed smooth function by this cutoff changes its L^2 norm and Dirichlet energy by terms tending to zero. Hence the puncture has zero H^1 capacity and the closed zero-flux form equals the full-plane free form. $\square$

The original physical effect belongs to Aharonov and Bohm [1]; the rigorous spectral and scattering treatment for general extension parameters is reviewed by Pankrashkin and Richard [2]. Their magnetic-potential sign corresponds to orders $|m + \alpha|$. Angular reflection, or $\alpha = -\beta$, translates to (3). The present formulas use one sign throughout. Reconstructing and testing this classical theory is the purpose of this package; classical formula discovery is not claimed.

2 Complete wave operators in the fixed sign convention

Put $\mu_m = |m|$ and $a_m = \pi(\mu_m - \nu_m)/2$. Define the free comparison by

$$W_{\pm} = \text{s-lim}_{t \rightarrow \pm\infty} e^{itH_{\beta}} e^{-itH_0}, \quad S = W_{+}^{*} W_{-}. \quad (6)$$

Theorem 2 (Wave operators and all scattering phases). *The limits in (6) exist and are onto. Their channel restrictions and the on-shell angular scattering phases are*

$$W_{\pm, m} = e^{\mp i a_m} \mathcal{J}_{\nu_m} \mathcal{J}_{\mu_m}, \quad S_m = e^{i\pi(|m| - |m - \beta|)}. \quad (7)$$

For $0 < \beta < 1$ this is $e^{i\pi\beta}$ on $m \geq 1$ and $e^{-i\pi\beta}$ on $m \leq 0$.

Proof. The real-order large-argument expansion is

$$J_{\nu}(kr) = \left(\frac{2}{\pi kr} \right)^{1/2} \cos(kr - \pi\nu/2 - \pi/4) + O((kr)^{-3/2}). \quad (8)$$

The incoming radial coefficient has phase $e^{i\pi\nu/2}$ and the outgoing one phase $e^{-i\pi\nu/2}$. Matching them to the free order μ_m gives e^{ia_m} for W_{-} and e^{-ia_m} for W_{+} . The rigorous dense-packet argument takes smooth spectral support in a compact subset of $k > 0$. Stationary phase puts the incoming packet at $r \approx 2|t|/k$ for negative time and the outgoing packet there for positive time. The remainder in (8) tends to zero in radial L^2 in that region, while integration by parts controls the nonstationary complement. This proves the channel limits on a dense set. The wave-operator theorem in [2, Lemma 7 and the original AB channel calculation] supplies the rigorous existence and completeness statement for exactly this regular extension. The explicit phase matching fixes its sign in our convention. Uniform unitary bounds extend the channel limits to all radial vectors; finite angular sums extend them to $\mathcal{H}$. Every Hankel product is onto, so their direct sum is onto. Multiplication of $W_{+, m}^{*}$ and $W_{-, m}$ gives the last formula. Splitting at $m = 0$ proves the two constant phases. $\square$

3 The forward distribution is part of scattering

Let P_{+} select $m \geq 1$ and P_{-} select $m \leq 0$ in $L^2(S^1, d\theta)$. Then $S = e^{i\pi\beta} P_{+} + e^{-i\pi\beta} P_{-}$. Write $\delta_{2\pi}$ for the periodic delta with integral one, and let θ be outgoing angle minus incoming angle.

Theorem 3 (Full kernel and away-forward cross section). *The convolution kernel relative to $d\theta$ is*

$$K_S(\theta) = \cos(\pi\beta) \delta_{2\pi}(\theta) - \frac{\sin(\pi\beta)}{2\pi} \text{PV} \cot(\theta/2) - \frac{i \sin(\pi\beta)}{2\pi}. \quad (9)$$

Under the amplitude convention $f = (2\pi i k)^{-1/2} \sum_m (S_m - 1) e^{im\theta}$, with $\sqrt{i} = e^{i\pi/4}$, away from $\theta = 0$ modulo 2π one has

$$f = -\frac{\sin(\pi\beta)}{\sqrt{2\pi i k}} (\cot(\theta/2) + i), \quad \frac{d\sigma}{d\theta} = \frac{\sin^2(\pi\beta)}{2\pi k \sin^2(\theta/2)}. \quad (10)$$

Proof. Introduce Abel weights $\rho^{|m|}$, $0 < \rho < 1$, in the geometric series. Their distributional limits are

$$\sum_{m \geq 1} e^{im\theta} = \pi \delta_{2\pi} - \frac{1}{2} + \frac{i}{2} \text{PV} \cot(\theta/2), \quad (11)$$

$$\sum_{m \leq 0} e^{im\theta} = \pi \delta_{2\pi} + \frac{1}{2} - \frac{i}{2} \text{PV} \cot(\theta/2). \quad (12)$$

Multiplying by the two channel phases and dividing by 2π gives (9). Subtracting the identity kernel retains $(\cos \pi\beta - 1) \delta_{2\pi}$. Off its support the amplitude equals the function in (10); $|\cot(\theta/2) + i|^2 = \csc^2(\theta/2)$ gives the cross section. No square of a distribution is taken. $\square$

For nonzero flux in the chosen interval the integrated away-forward cross section diverges. The two angular projectors nevertheless show S is exactly unitary. These statements concern different objects and are consistent.

4 Local heat kernel and global noncompactness

For $t > 0$ write $z = rr'/(2t)$. Weber's exponential integral [3, 10.22.67] and the angular normalizer give the kernel relative to $r' dr' d\theta'$:

$$K_\beta(t; r, \theta; r', \theta') = \frac{e^{-(r^2+r'^2)/(4t)}}{4\pi t} \sum_{m \in \mathbb{Z}} e^{im(\theta-\theta')} I_{|m-\beta|}(z). \quad (13)$$

Indeed each radial term before evaluation is

$$\int_0^\infty e^{-tk^2} J_\nu(kr) J_\nu(kr') k dk = \frac{e^{-(r^2+r'^2)/(4t)}}{2t} I_\nu(z). \quad (14)$$

Lemma 2 (An explicit angular tail). *For $z > 0$, $\nu_0 \geq 0$, and $z < 2(\nu_0 + 1)$, put*

$$B(\nu_0, z) = \frac{e^{z^2/4} (z/2)^{\nu_0}}{\Gamma(\nu_0 + 1) [1 - z/(2(\nu_0 + 1))]} \quad (15)$$

The absolute tail of the sum in (13) outside $|m| \leq M$ is at most $B(M + 1 - \beta, z) + B(M + 1 + \beta, z)$ whenever both denominators are positive. Consequently the series converges locally uniformly for positive t, r, r' .

Proof. The positive defining series of I_ν and $(\nu + 1)_j \geq j!$ imply $I_\nu(z) \leq (z/2)^\nu e^{z^2/4} / \Gamma(\nu + 1)$ for $\nu \geq 0$. Consecutive terms of this upper bound have ratio $z/[2(\nu + 1)]$. Starting at ν_0 , sum the geometric majorant to get $\sum_{j \geq 0} I_{\nu_0+j}(z) \leq B(\nu_0, z)$. The positive and negative omitted angular orders start at $M + 1 - \beta$ and $M + 1 + \beta$ respectively. This proves the stated bound; on compact parameter sets one can take M uniformly large. At $z = 0$ the series limits apply. $\square$

The free-flux identity $\sum_m e^{im\phi} I_{|m|}(z) = e^{z \cos \phi}$ reduces (13) to the ordinary free plane Gaussian.

Theorem 4 (Global heat and scattering determinant obstructions). *For any flux and $t > 0$, e^{-tH_β} is noncompact and has infinite positive extended trace. For $0 < \beta < 1$, $S - I$ is noncompact, so the ordinary trace-class Fredholm determinant $\det(I + (S - I))$ is unavailable. In addition,*

$$\prod_{m=-M}^M S_m = e^{-i\pi\beta}, \quad \prod_{m=-M}^{M+1} S_m = 1 \quad (M \geq 0). \quad (16)$$

Proof. In a single Hankel channel, the heat semigroup is multiplication by e^{-tk^2} . On any fixed positive finite interval of k this function is bounded below. Normalized indicators of disjoint positive-measure subsets form an orthonormal sequence whose images have disjoint supports and norms bounded below, excluding compactness. A positive trace-class operator is compact, so the positive extended trace is infinite.

For the scattering operator, the angular orthonormal vectors satisfy $\|(S - I)e_m\| = 2 \sin(\pi\beta/2) > 0$ for every m ; their images remain orthogonal. The same compactness criterion applies. In the first window of (16), there are M positive and $M + 1$ nonpositive indices; the second has $M + 1$ of each. The two products follow. Both windows exhaust all integers, so symmetric-cutoff stability by itself does not supply an exhaustion-independent determinant. $\square$

This result addresses ordinary determinants and traces. A separately specified relative or renormalized object would require its own definition and theorem.

5 Gauge covariance and the precise time-reversal question

For integer n , let $U_n f = e^{in\theta} f$ and let K denote position-space conjugation. Directly on the form core and then by closure,

$$H_{\beta+n} = U_n H_\beta U_n^{-1}, \quad K H_\beta K = H_{-\beta}. \quad (17)$$

Noninteger U_n would not preserve single-valuedness on this Hilbert space.

Theorem 5 (Position-preserving physical time reversal). *Restrict physical time reversal to $T_n = U_n K$, with no spatial reflection: it fixes position and reverses mechanical momentum when it is a symmetry. It preserves H_β exactly when $n = 2\beta \in \mathbb{Z}$. In the fundamental interval, these fluxes are 0 and $1/2$, and $T_n^2 = I$.*

Proof. Equation (17) gives $T_n H_\beta T_n^{-1} = H_{n-\beta}$. Equality of the coefficients of $-i\partial_\theta$ in the radial differential expressions forces $n - \beta = \beta$. Conversely that equality gives form and domain invariance. Since $K U_n K = U_{-n}$, the square is one. Under $n = 2\beta$, the mechanical angular momentum is also reversed: $T_n(-i\partial_\theta - \beta) T_n^{-1} = -(-i\partial_\theta - \beta)$. $\square$

The reflection $Rf(r, \theta) = f(r, -\theta)$ sends H_β to $H_{-\beta}$, so RK is a commuting antiunitary for every flux. It moves spatial points; its existence does not establish the position-preserving physical symmetry just classified. Abstract spectral antiunitaries cannot replace that test.

6 Evidence, provenance, and the Route-A endpoint

The frozen receipt contains 520 exact channel rows, 680 integer-gauge rows, eight physical-symmetry rows, and 520 cutoff rows. Six high-precision Weber kernels are reconstructed independently by integration of the ordinary-Bessel spectral formula. Twelve away-forward cross sections are reconstructed from the complex amplitude. These finite checks are numerical regression where decimals occur; they are not interval certification or proofs of completeness. The separate proof document records the analytic dependencies.

The package's substantive increment is the connected audit of the Friedrichs domain, all channel phases, full distribution, local heat tail, global noncompactness, cutoff failure, and physical symmetry. Classical AB formulas remain attributed to their original literature. The nearest local magnetic owners concern compact magnetic tori, monopole spheres, and Grushin cylinders; none is this noncompact punctured-plane Friedrichs scattering problem.

Off the puncture the classical magnetic field vanishes. Mechanical momentum is therefore constant along every nonsingular trajectory of positive energy, which is a straight line and has no periodic return. A trajectory hitting the puncture is incomplete here; zero-energy stationary continua are not isolated primitive cycles. Angular integers and winding phases supply no rational-prime carrier. Free flux, neighboring fluxes, and angular-label controls preserve this failure. The strict tuple is

$$(A0_{\text{fail}}, A1_{\text{fail}}, A2_{\text{fail}}, A3_{\text{fail}}, A4_{\text{unitary/scattering}}), \quad \text{Route A rejected.} \quad (18)$$

The scope is NO_BAD_EULER_OR_ROOT_NUMBER. No target arithmetic local data, target Euler factors, root number, automorphy, target divisor or counting law, target functional equation, target-zero match, Hilbert–Pólya operator, or Route B is claimed. The natural source scattering Hamiltonian earns A4 alone and does not repair the failed earlier gates.

References

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